

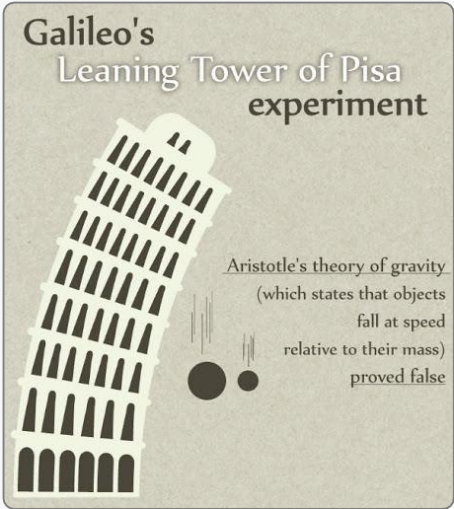
Modelica文本一般结构及语法概览

Modelica语法详解

苏州同元软控信息技术有限公司

2024年2月27日

先来看两个现象



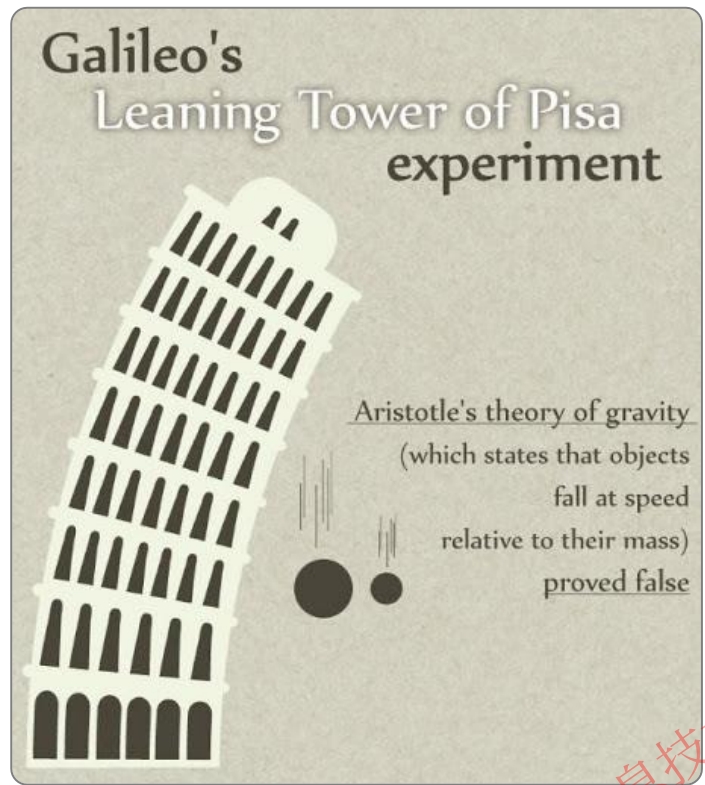
自由落体



弹跳小球

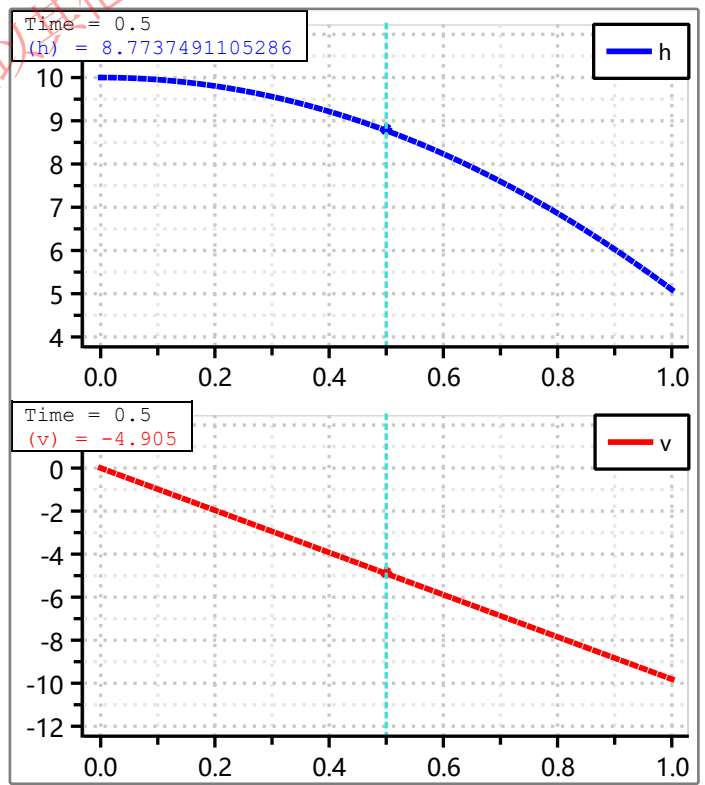
问题：如何对以上物理现象进行仿真分析？

自由落体

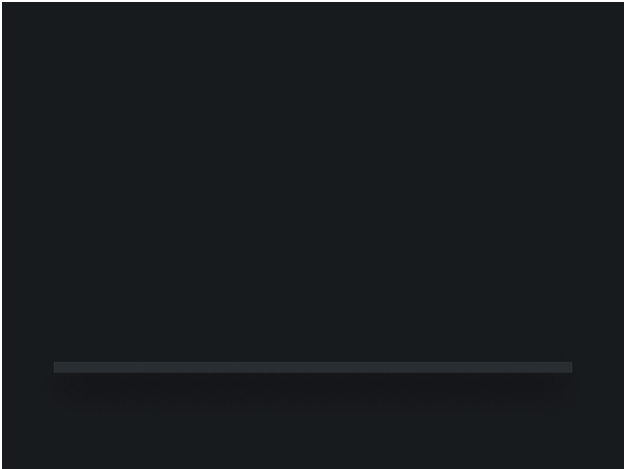


$$\begin{cases} v = \dot{h} \\ a = \dot{v} = -g \end{cases}$$

```
class Fall "自由落体"  
  Real g = 9.81;  
  Real h(start = 10);  
  Real v;  
  equation  
    der(h) = -v;  
    der(v) = g;  
end Fall;
```

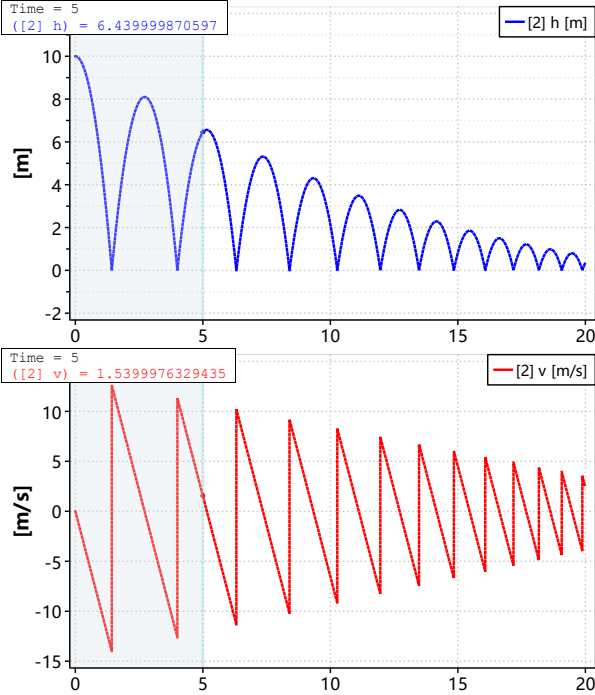


弹跳小球



在自由落体的基础上，小球在触地瞬间，重置速度即可

```
class Ball "弹跳小球"  
  constant Modelica.SIunits.Acceleration g = 9.8 "重力加速度";  
  parameter Real coef = 0.9 "弹性系数";  
  parameter Modelica.SIunits.Height h0 = 10 "初始高度";  
  Modelica.SIunits.Height h(start = h0) "小球高度";  
  Modelica.SIunits.Velocity v "小球速度";  
  Boolean flying "是否运动";  
equation  
  flying = not (h <= 0 and v <= 0);  
  der(v) = if flying then -g else 0;  
  v = der(h);  
  when h <= 0 then  
    reinit(v, -coef * v);  
  end when;  
end Ball;
```



Modelica代码包含哪些要素？ 如何构建完备的Modelica代码？

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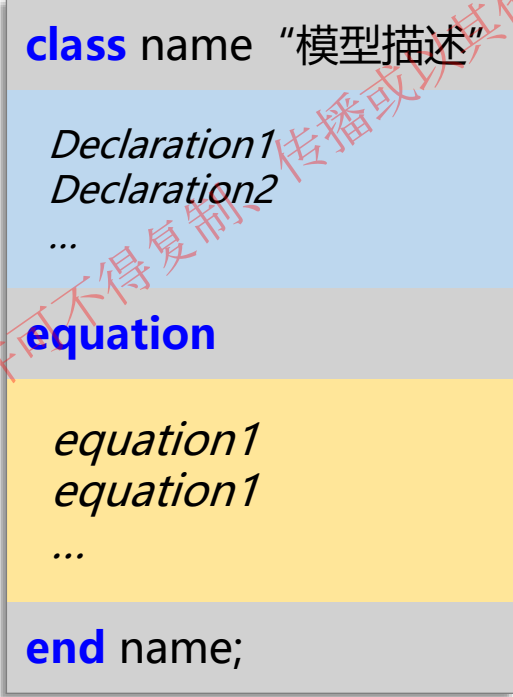
2. Modelica语法概览

1. Modelica文本一般结构

两个模型代码有什么共同之处？

```
class Fall "自由落体"  
  Real g = 9.81;  
  Real h(start = 10);  
  Real v;  
equation  
  der(h) = -v;  
  der(v) = g;  
end Fall;
```

```
class Ball "弹跳小球"  
  final parameter Modelica.SIunits.Acceleration g = 9.8 "重力加速度";  
  parameter Real coef = 0.9 "弹性系数";  
  parameter Modelica.SIunits.Height h0 = 10 "初始高度";  
  Modelica.SIunits.Height h (start = h0) "小球高度";  
  Modelica.SIunits.Velocity v "小球速度";  
  Boolean flying "是否运动";  
equation  
  flying = not (h <= 0 and v <= 0);  
  der(v) = if flying then -g else 0;  
  v = der(h);  
  when h <= 0 then  
    reinit(v, -coef * v);  
  end when;  
end Ball;
```



开始模型定义: class+模型名称+ "模型描述"

- 声明模型参数
- 声明模型变量
- 声明继承或被调用的模型 (如接口)

- 通过方程或算法, 描述模型行为

结束模型定义: end+模型名称;



1.1 Modelica文本一般结构-模型的开始与结束

```
class name "模型描述"
...
end name;
```

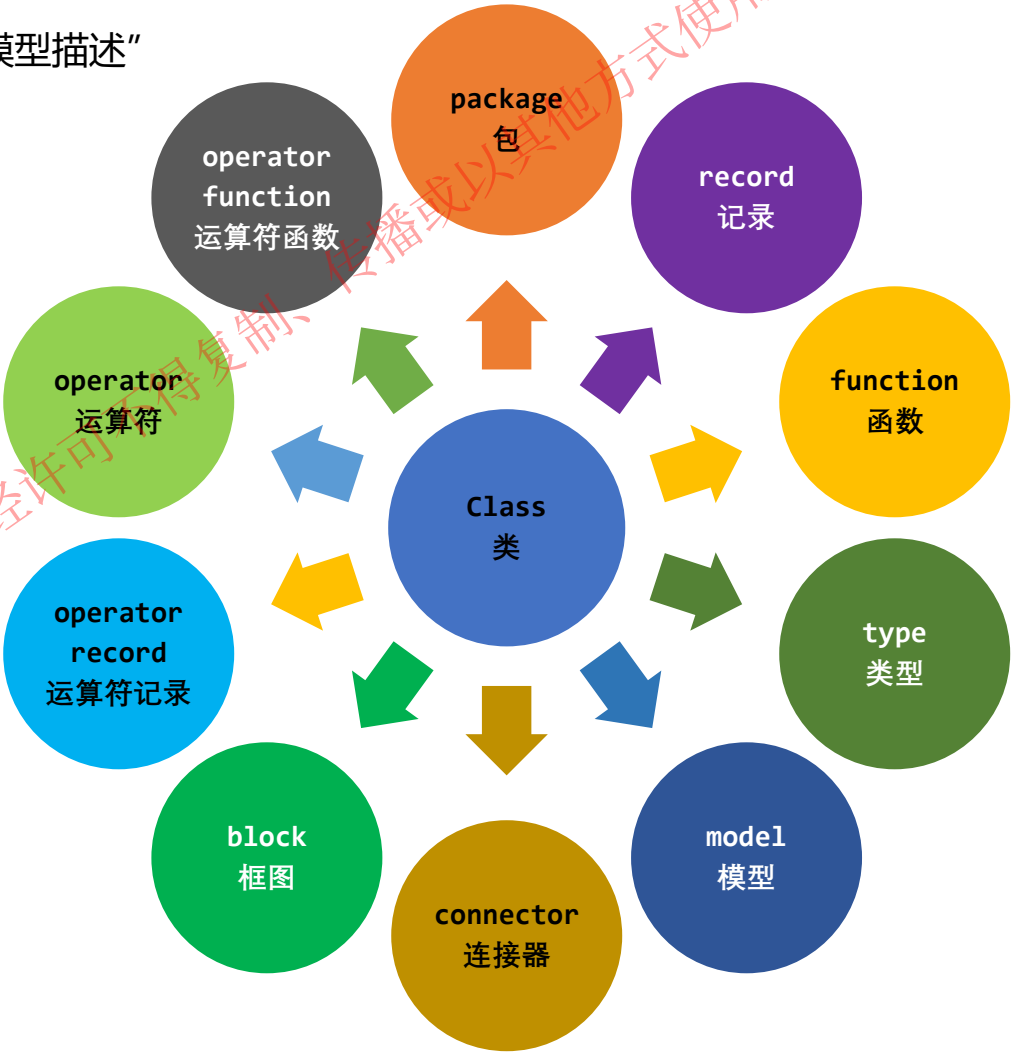
开始模型定义: 类型+模型名称+ “模型描述”
• ...
结束模型定义: end+模型名称;

类型是对模型基本属性的描述
Modelica语言定义了10种特化类

其他类型定义

```
model Example1 "示例1"
...
end Example1;
```

```
connector Example2 "示例2"
...
end Example2;
```



1.2 Modelica文本一般结构-变量声明

class name “模型描述”

Variable Declarations

...

end name;

开始模型定义: 类型+模型名称+ “模型描述”

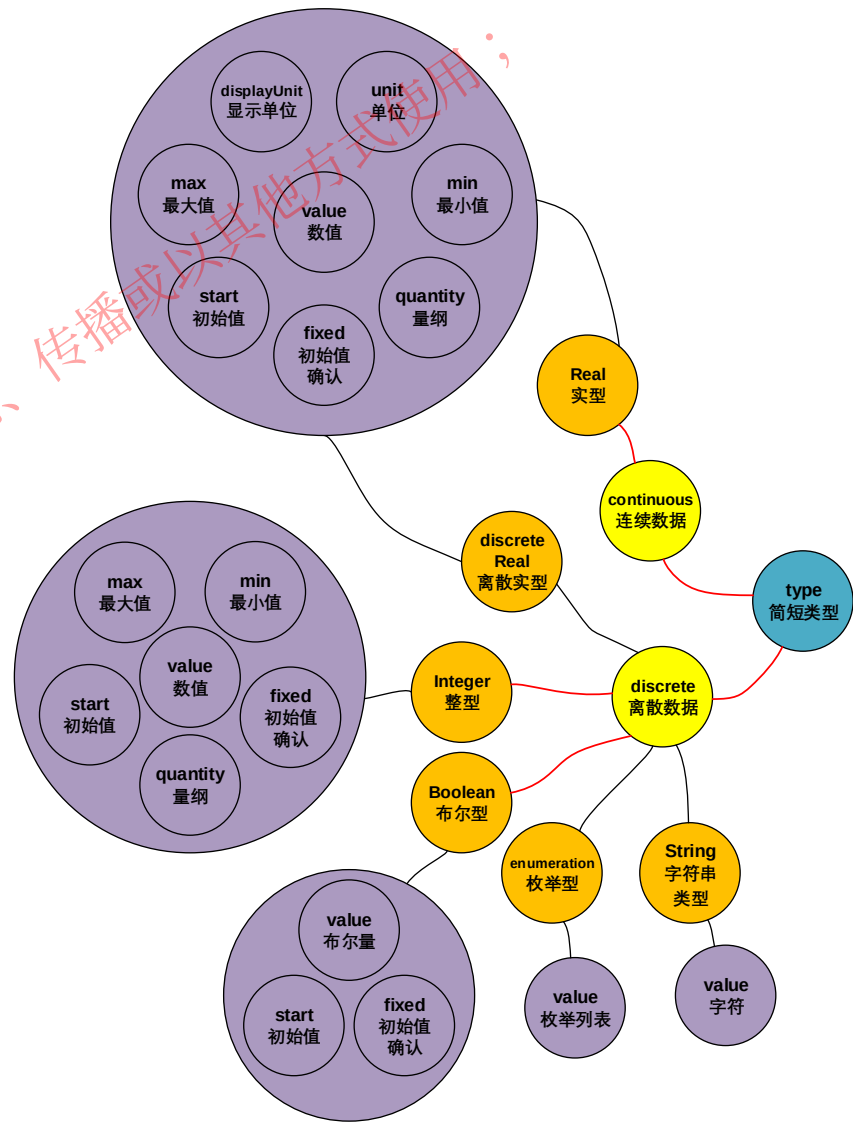
- 变量声明
- ...

结束模型定义: end+模型名称;

变量定义一般格式:
(数据前缀+)**数据类型**+变量名称 (属性描述) + “变量描述”
例如: **Modelica.SIunits. Height** h0 (start=10) “初始高度”;

Modelica语言定义了5种数据类型以及多种数据属性关键词

```
class VariableDeclaration “变量声明示例”  
  Real A(max=1) “A变量描述”;  
  Integer B(min=0) “B变量描述”;  
  Boolean C(start=true) “C变量描述”;  
  String D “D变量描述”;  
  type E = enumeration(e1 , e2 , ... , en) “E变量描述” ;  
  ...  
end VariableDeclaration;
```



1.3 Modelica文本一般结构-变量声明

`class` name “模型描述”

`Parameter Declarations`
`Variable Declarations`

...

`end` name;

开始模型定义: 类型+模型名称+ “模型描述”

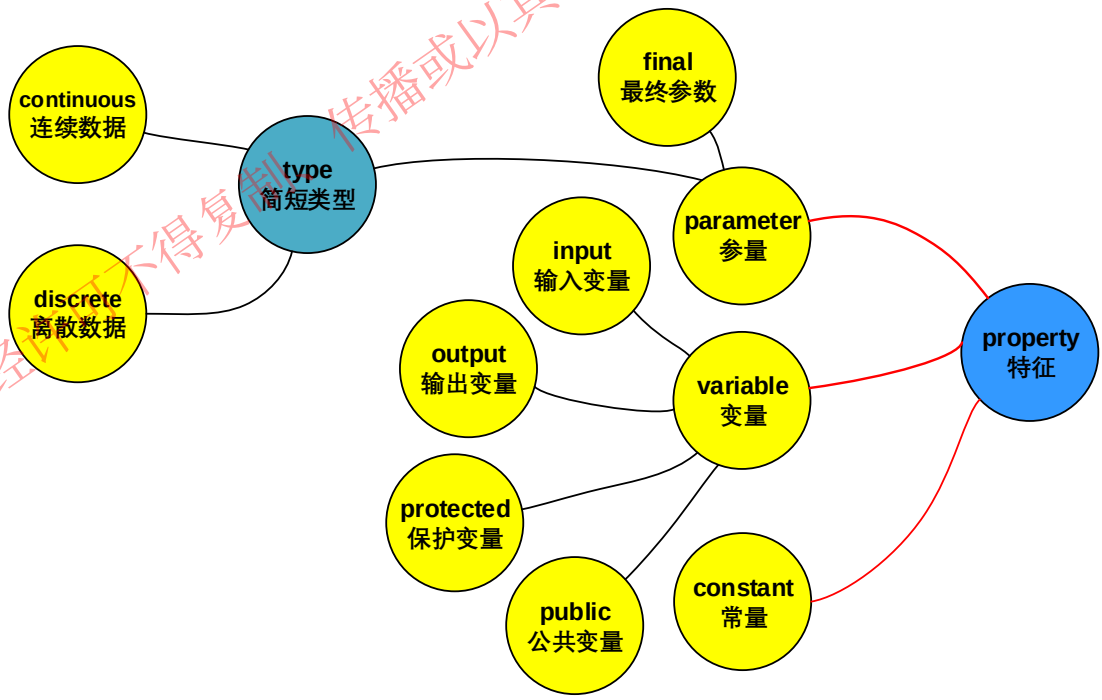
- 参数声明
- 变量声明
- ...

结束模型定义: end+模型名称;

参数声明也是一种变量声明，一般格式如下：
数据前缀+数据类型（单位）=值+ “参数描述”

Modelica语言定义了4类8个数据前缀

```
class ParameterDeclaration “参数声明示例”  
  parameter Integer A=1 “A参数描述”;  
  constant Real B(unit= “m/s” )=1 “B参数描述”;  
  final parameter Real C=1 “C参数描述”;  
end ParameterDeclaration;
```



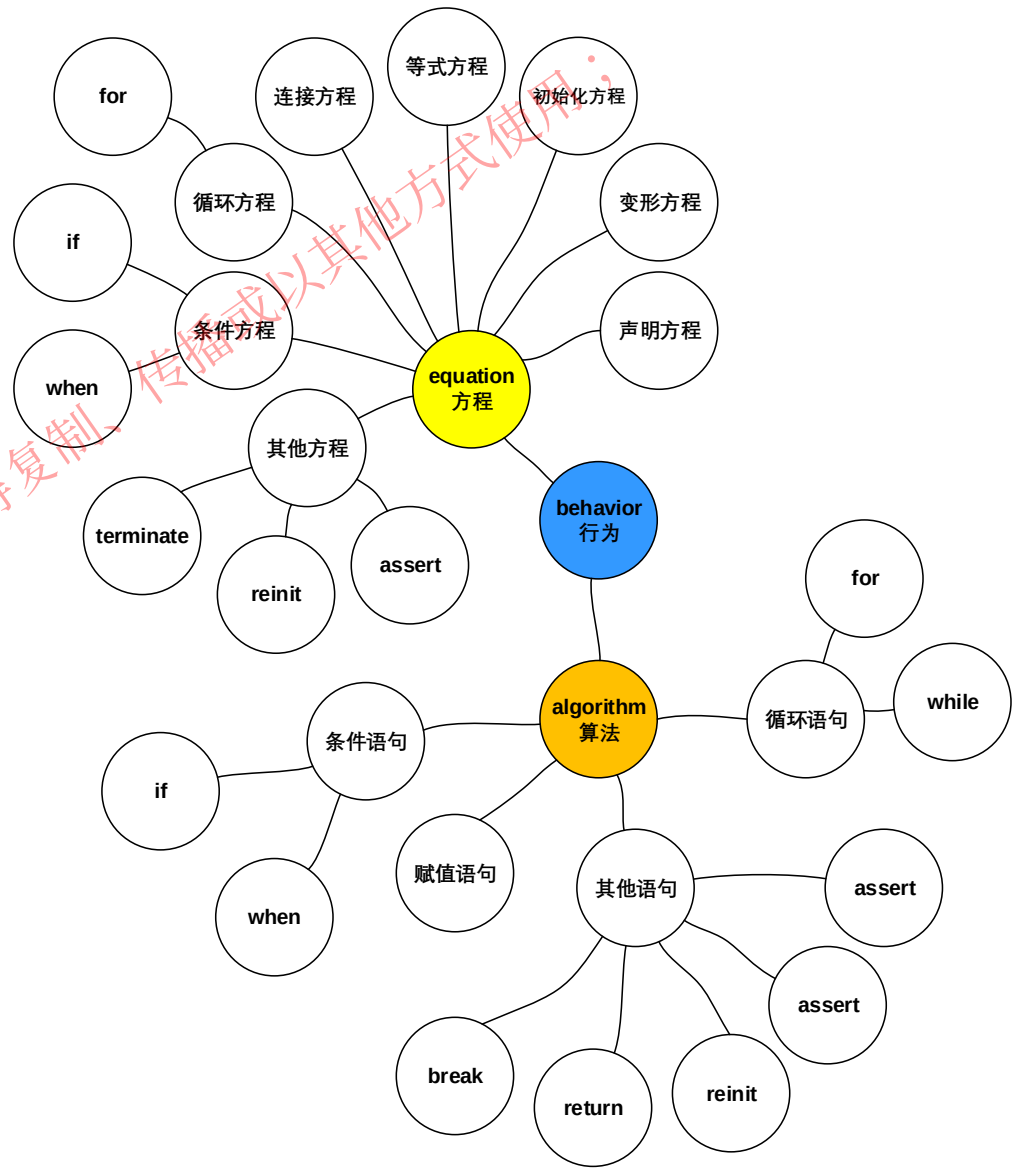
1.4 Modelica文本一般结构-模型行为描述

class name “模型描述”	开始模型定义: 类型+模型名称+ “模型描述”
Parameter Declarations	• 参数声明
Variable Declarations	• 变量声明
algorithm	模型行为描述
...	• 算法
equation	模型行为描述
...	• 方程
end name;	结束模型定义: end+模型名称;

Modelica语言支持使用**方程**或**算法**对模型行为进行描述，
对应的关键词分别为**equation**与**algorithm**。

```
class Example “模型行为描述示例”  
  parameter Real a=1;  
  Real b;  
  Real c;  
  equation  
    a=b+c;  
    b=1;  
end Example;
```

完整模型一般结构

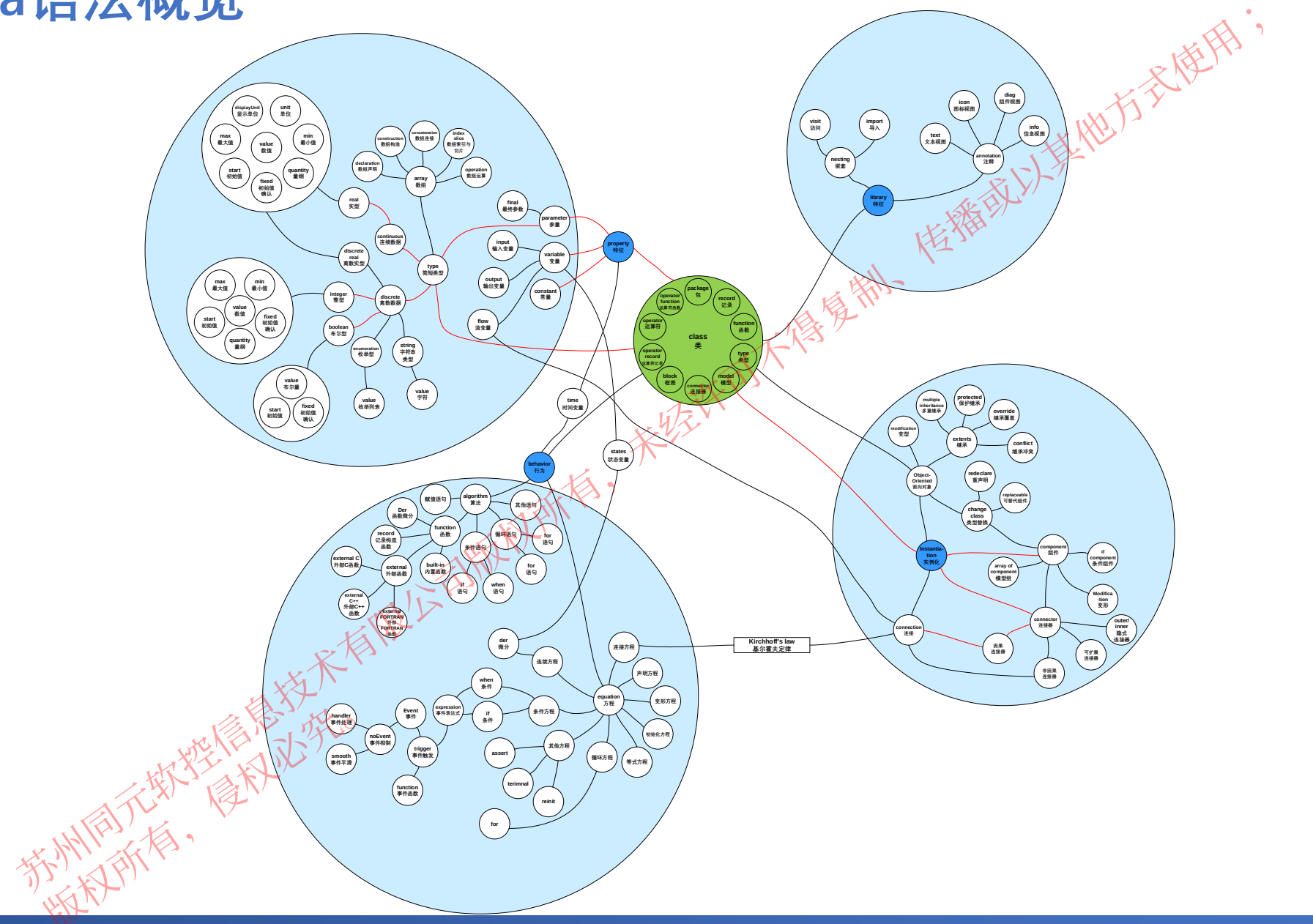


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1. Modelica文本一般结构

2. Modelica语法概览

2. Modelica语法概览



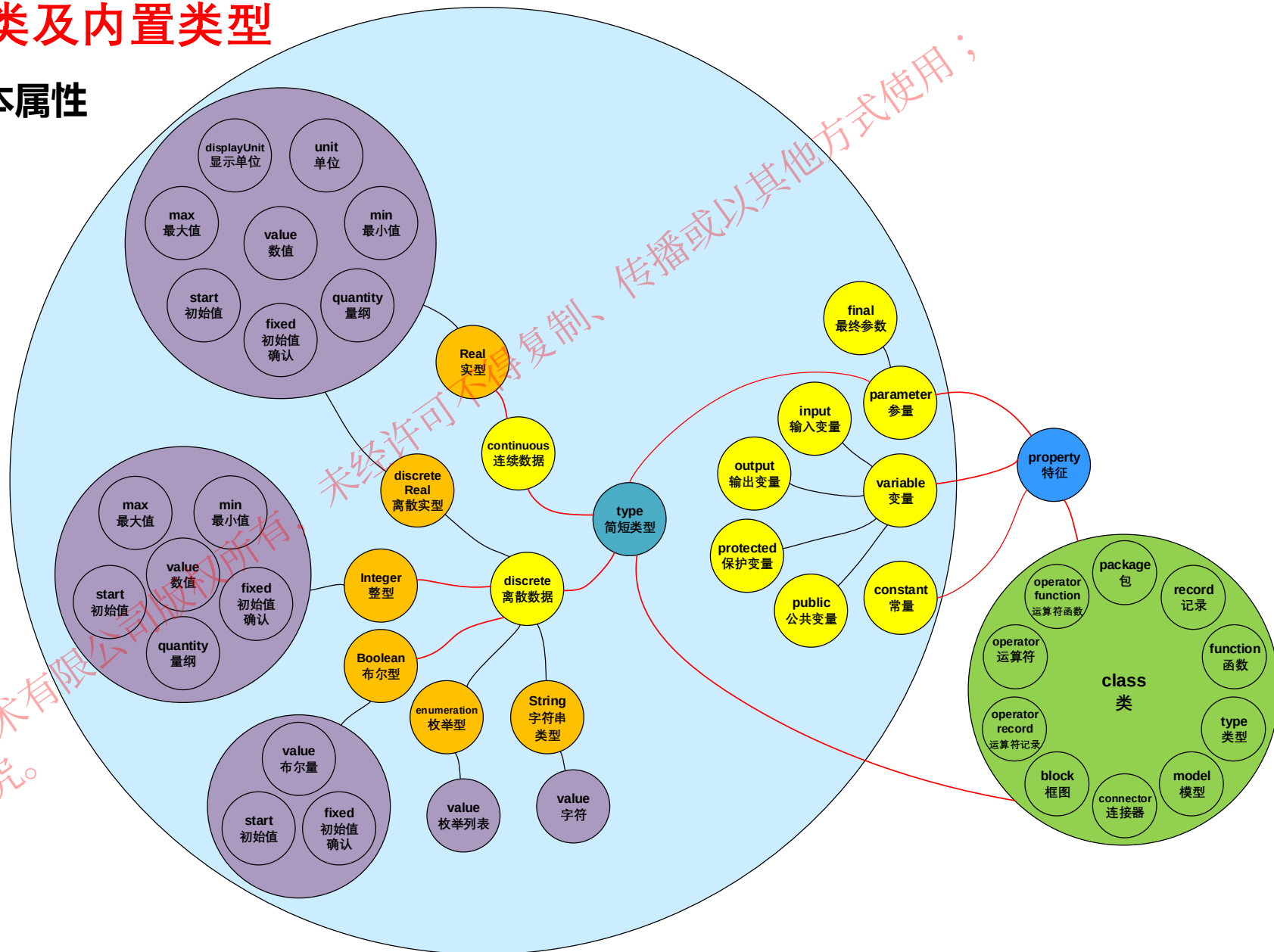
2.1 Modelica语法概览-类及内置类型

类及内置类型——构成模型的最基本属性

Modelica语言定义了:

- 10种特化类
- 5种数据类型
- 多种前缀关键词
- 多种数据属性关键词

```
model Fall "自由落体"  
  final parameter Real g = 9.81;  
  Real h(start = 10);  
  Real v;  
equation  
  der(h) = v;  
  der(v) = -g;  
end Fall;
```



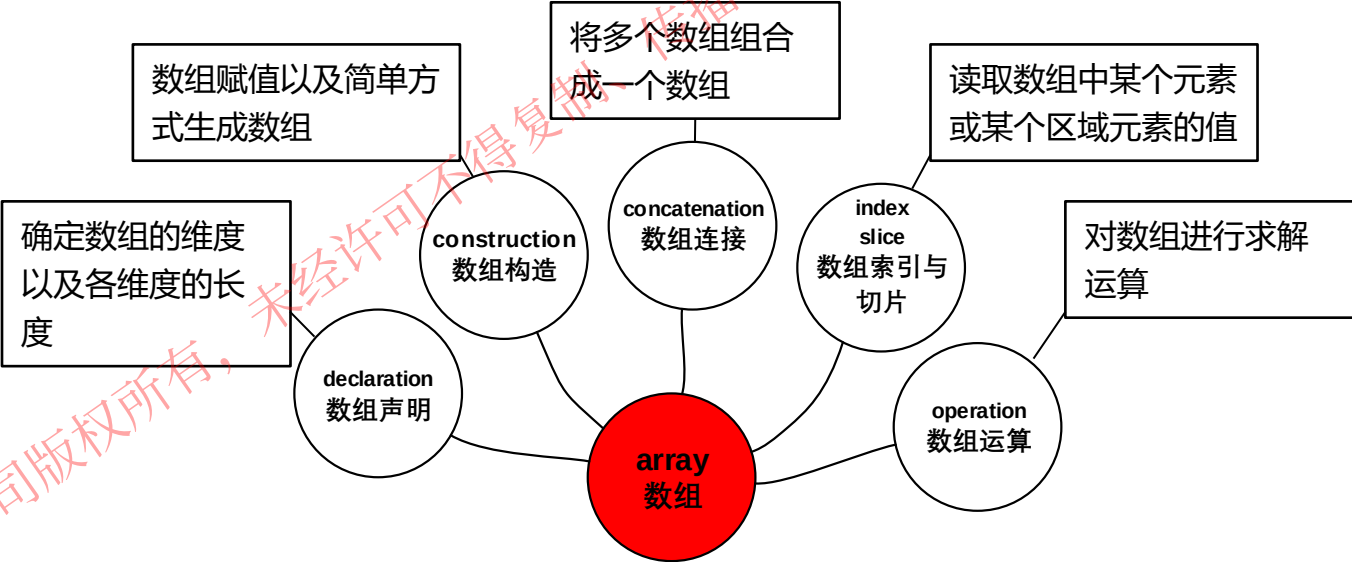
2.2 Modelica语法概览-数组

数组—多个同类型变量的集合

Modelica可定义多维数组：

- 0维数组：标量，如：3.14
- 1维数组：向量，如：
- 2维数组：矩阵，如： $\begin{Bmatrix} v_x & v_y & v_z \\ 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{Bmatrix}$

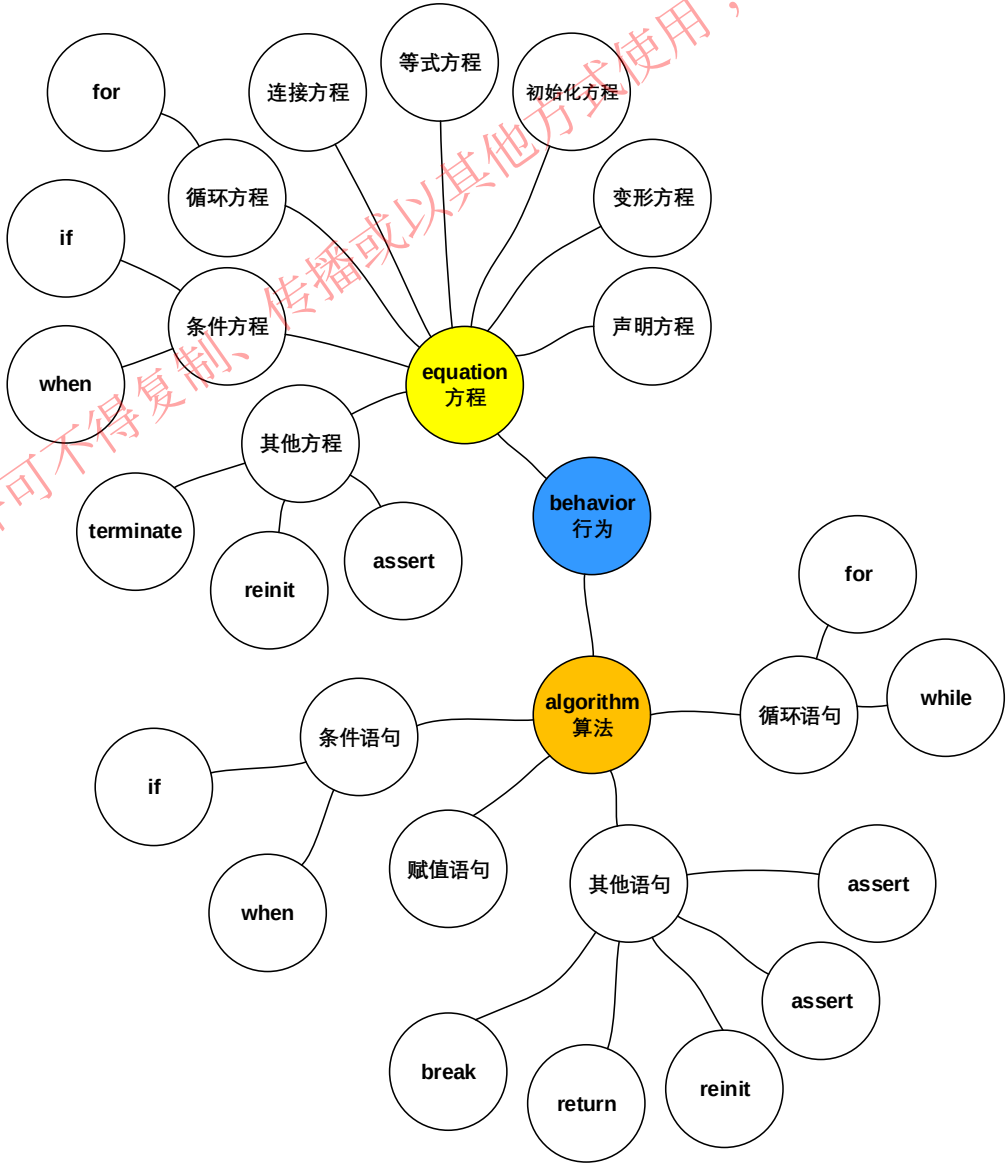
```
.....
model FlatThrow "平抛运动"
  final parameter Real g = 9.81;
  parameter Real v0 = 1;
  Real x(start = 0);
  Real h(start = 10);
  Real position[2] = {x, h};
  Real v[2];
equation
  v[1] = v0;
  v[1] = der(x);
  v[2] = der(h);
  der(v[2]) = -g;
end FlatThrow;
```



2.3 Modelica语法概览-模型行为描述

模型行为描述—通过方程或算法定义模型的行为

声明区域方程	声明方程		
	变形方程		
	初始化方程	initial equation	
方程区域方程	常规方程	等式方程	=
		连接方程	connect
		循环方程	for
		条件方程	if
			when
		其他方程	assert()
			terminate
			reinit()
算法区语句	算法语句	赋值语句	“:=”
		循环语句	for
			while
		条件语句	if
			when
			break
			return
		其他语句	assert
			terminate
			reinit

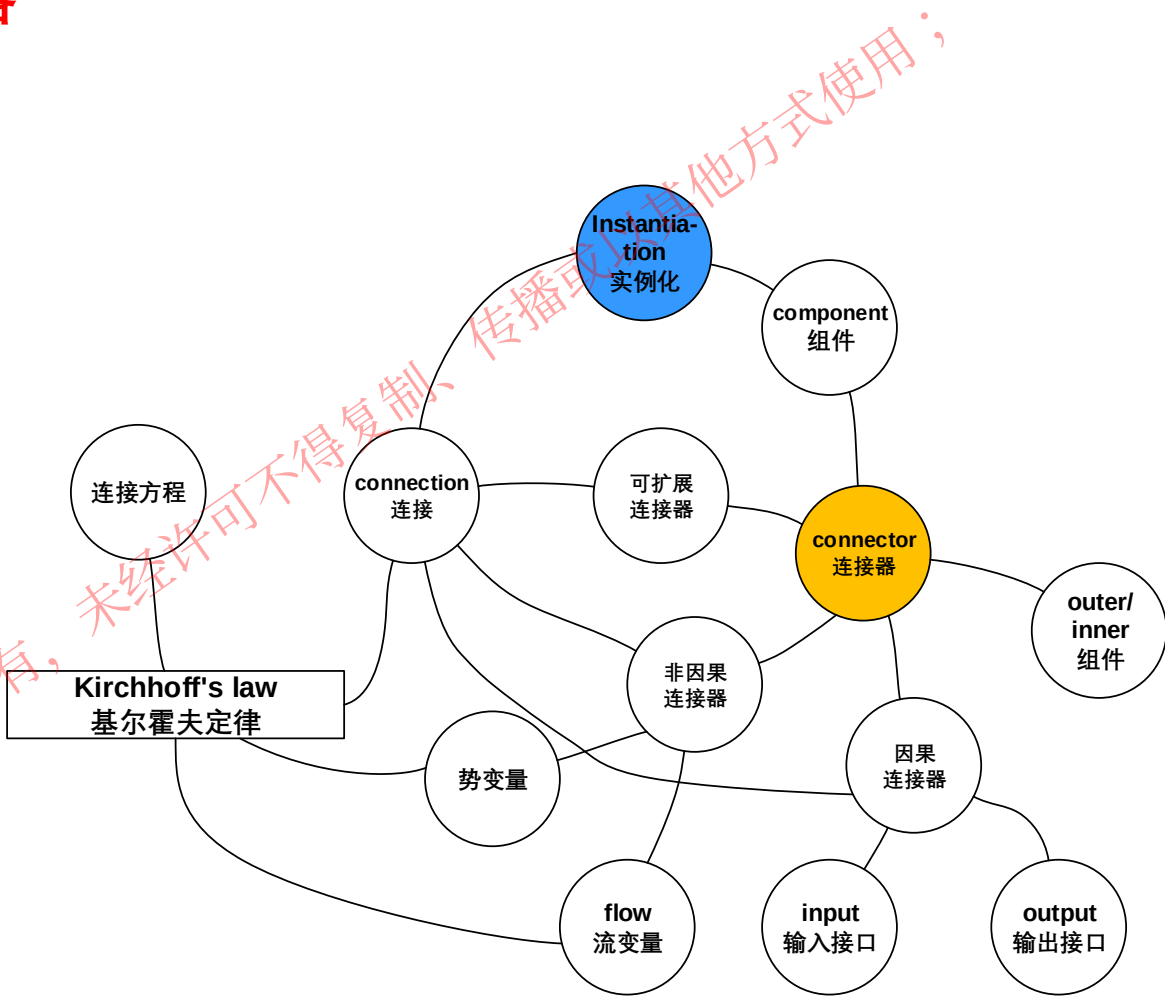
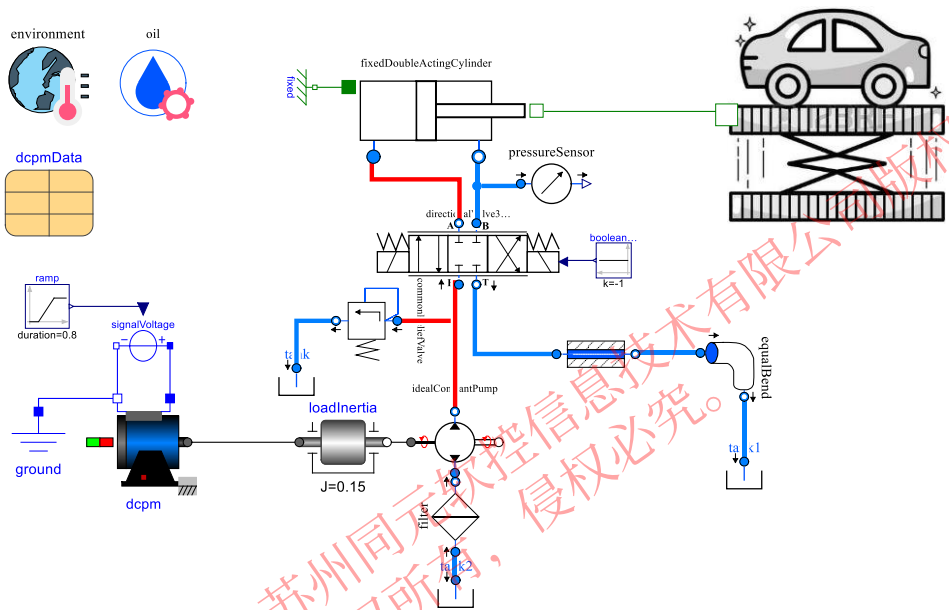


2.4 Modelica语法概览-连接与连接器

连接与连接器一定义组件与外部的通讯

连接器：

- 因果连接器
- 非因果连接器
- 可扩展连接器
- 隐式连接器

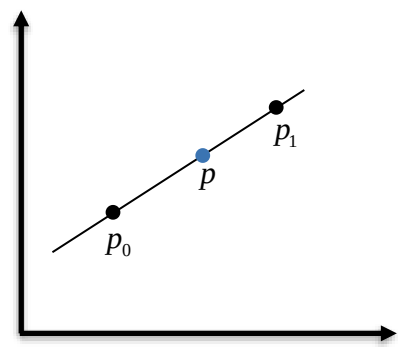


2.5 Modelica语法概览-函数

函数—数学模型的天然组成部分

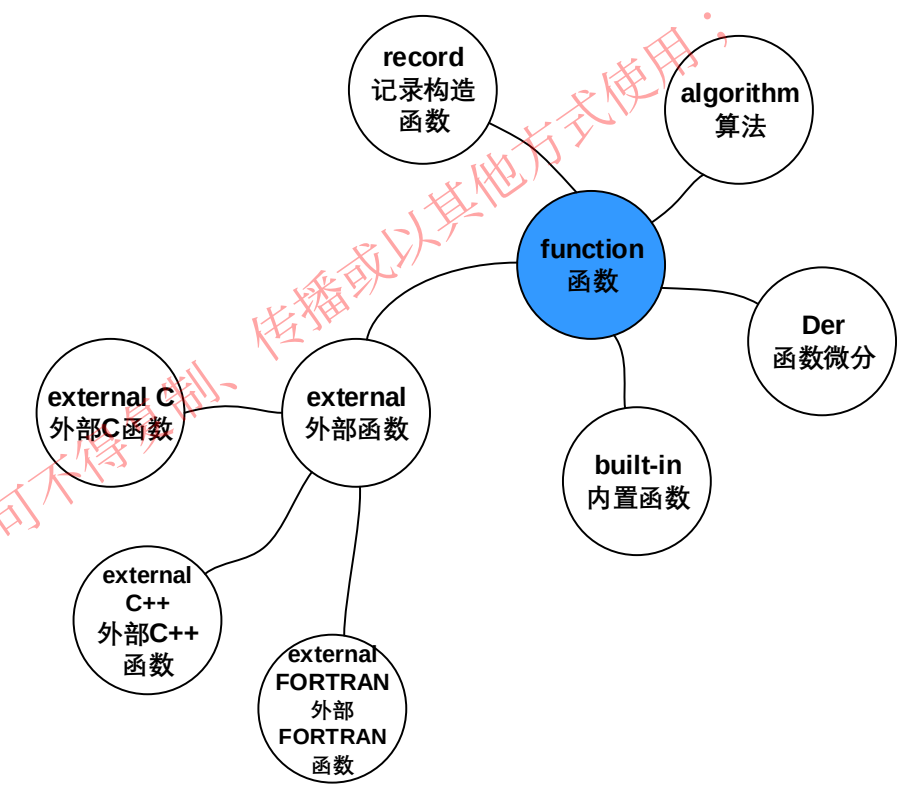
Modelica提供了：

- 多种内部函数
- 自定义函数
- C、C++、FORTRAN外部函数集成



如何计算一条直线上的点？

$$y(x, x_0, y_0, x_1, y_1) = x \frac{y_1 - y_0}{x_1 - x_0} - \frac{x_1 + x_0}{2} \frac{y_1 - y_0}{2} + \frac{y_1 + y_0}{2}$$



```
function LineWithProtected
input Real x;
input Real p0[2];
input Real p1[2];
output Real y;
protected
Real m = (p1[2] - p0[2]) / (p1[1] - p0[1]);
Real b = (p1[2] + p0[2] - m * (p1[1] + p0[1])) / 2.0;
algorithm
y := m * x + b;
end LineWithProtected;
```

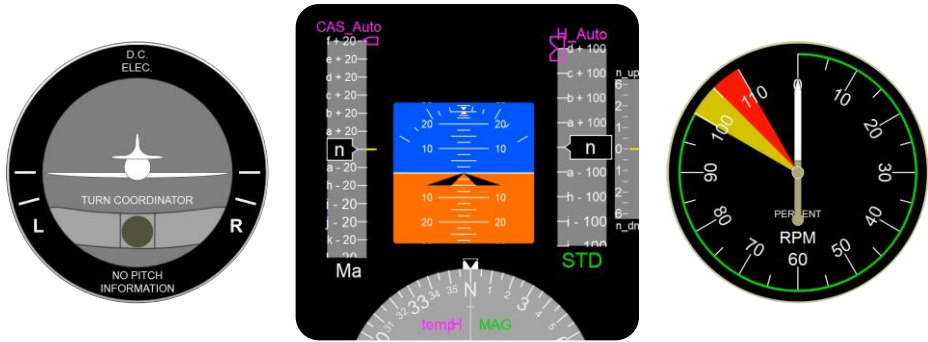
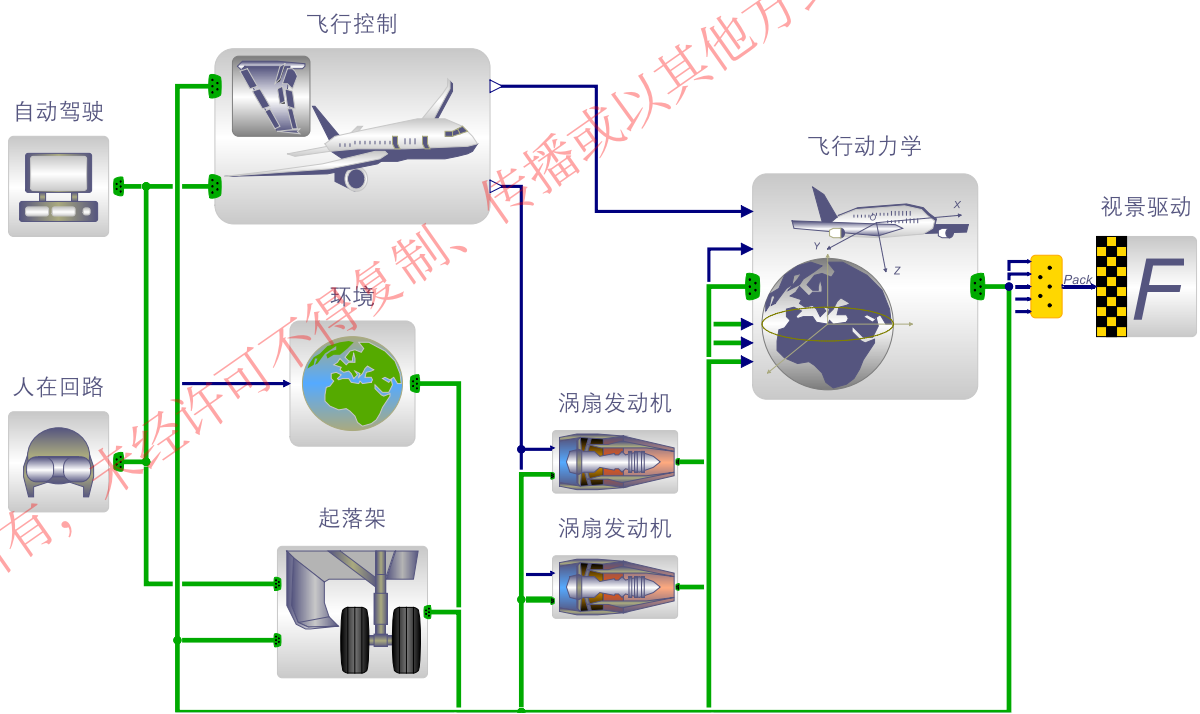
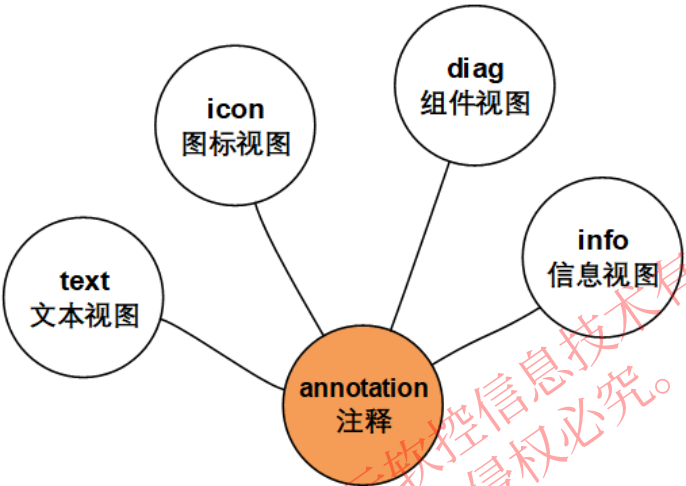


2.6 Modelica语法概览-注解

注解—定义与Modelica模型关联的附加信息

注解功能：

- 组件参数框设计
- 图标设计
- 模型帮助文档设计
- 动态显示设计
- 文本其他注释

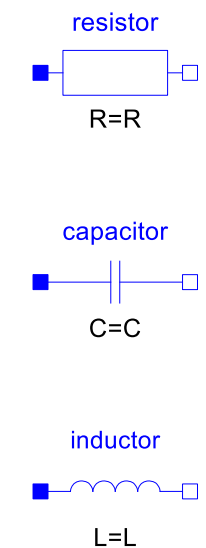


2.7 Modelica语法概览-模型重用

模型重用—极大地提高建模效率

模型重用:

- 继承重用
- 实例化重用
- 重声明重用



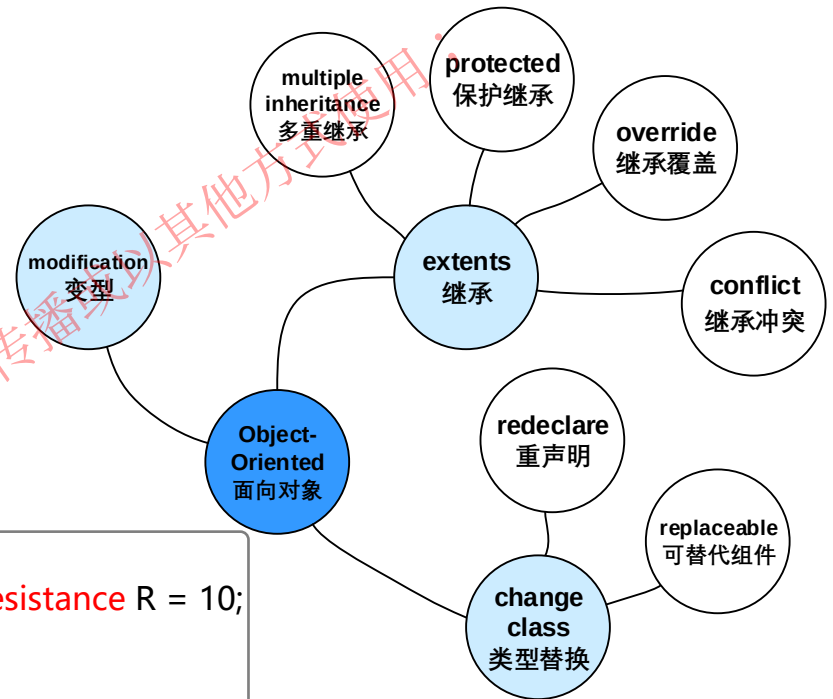
```
partial model PartialPort
  Modelica.SIunits.Voltage v;
  Modelica.SIunits.Current i;
  Interfaces.PositivePin p;
  Interfaces.PositivePin n;
equation
  v = p.v - n.v;
  i = p.i;
  -i = n.i;
end PartialPort;
```

基类

```
model Resistance
  parameter Modelica.SIunits.Resistance R = 10;
  extends PartialPort;
equation
  v = R * i;
end Resistance;

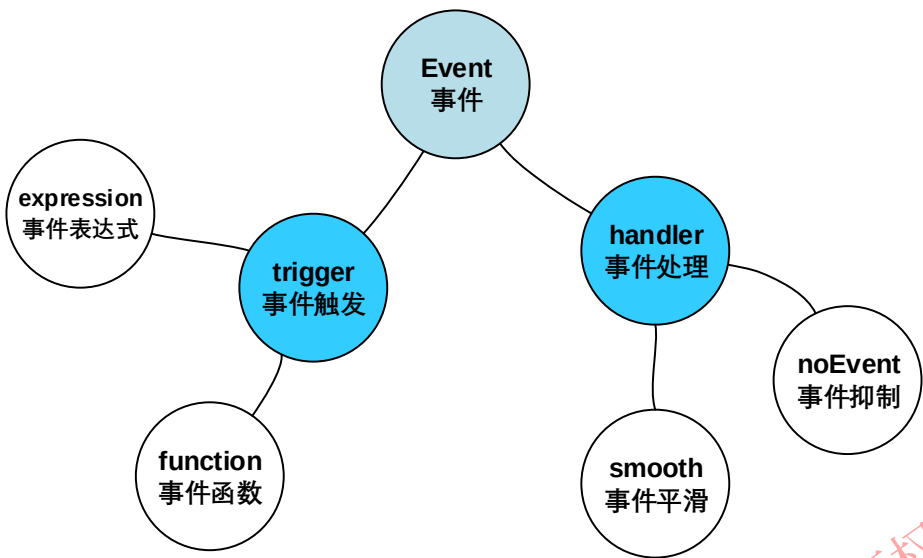
model Capacitor
  parameter Modelica.SIunits.Capacitance C = 1;
  extends PartialPort;
equation
  i = C * der(v);
end Capacitor;

model Inductor
  parameter Modelica.SIunits.Inductance L = 1;
  extends PartialPort;
equation
  L * der(i) = v;
end Inductor;
```



2.8 Modelica语法概览-事件

事件—系统变量在特定时间点上瞬时的、不连续的发生变化



```
model Ball "弹跳小球"
final parameter Real g = 9.8 "重力加速度";
parameter Real coef = 0.9 "弹性系数";
parameter Real h0 = 10 "初始高度";
Real h "小球高度";
Real v "小球速度";
Boolean flying "是否运动";
initial equation
h = h0;
equation
flying = not (h <= 0 and v <= 0);
der(v) = if flying then -g else 0;
v = der(h);
when h <= 0 then
  reinit(v, -coef * v);
end when;
end Ball;
```

输出

建模 仿真

Zero-Crossing of the 1st relational expression detected.
Time = 6.314285647, updating discrete events ...
Expression h<=0 became false, (h) - (0) = 1.000005537022688e-10
Expression h<=0 became false, (h) - (0) = 1.000005537022688e-10
Time = 6.314285647, event updating finished.

Zero-Crossing of the 1th relational expression detected.
Time = 8.397142853, updating discrete events ...
Expression h<=0 became true, (h) - (0) = -1.000178828647336e-10
Expression v<=0 became true, (v) - (0) = -10.20600049959137
Expression h<=0 became true, (h) - (0) = -1.000178828647336e-10
Iterating again to find consistent conditions ...
Expression v<=0 became false, (v) - (0) = 9.185400449632231
Time = 8.397142853, event updating finished.

Zero-Crossing of the 1th relational expression detected.
Time = 8.397142853, updating discrete events ...
Expression h<=0 became false, (h) - (0) = 1.000133511808675e-10
Expression h<=0 became false, (h) - (0) = 1.000133511808675e-10
Time = 8.397142853, event updating finished.

Simulation finished at time = 10 (stopTime = 10)
CPU time for Integration: 0.001s
Number of Time Events: 0
Number of State Events: 8
Number of Step Events: 0
Number of Grid Points: 500
Minimum integration stepsize: 1e-06
Maximum integration stepsize: 1.22803



建立知识规范，营造协同生态
积累工业模型，发展可控平台
融入中国创新，打造先进软件

谢谢！